Gradient descent: from direction to progress

Generated 10/31/2026 · AI notes, summary & key points

SUMMARY

Follow the gradient

Gradient descent improves a model by repeatedly updating its parameters to reduce a loss function. Each step combines a direction with a learning rate.

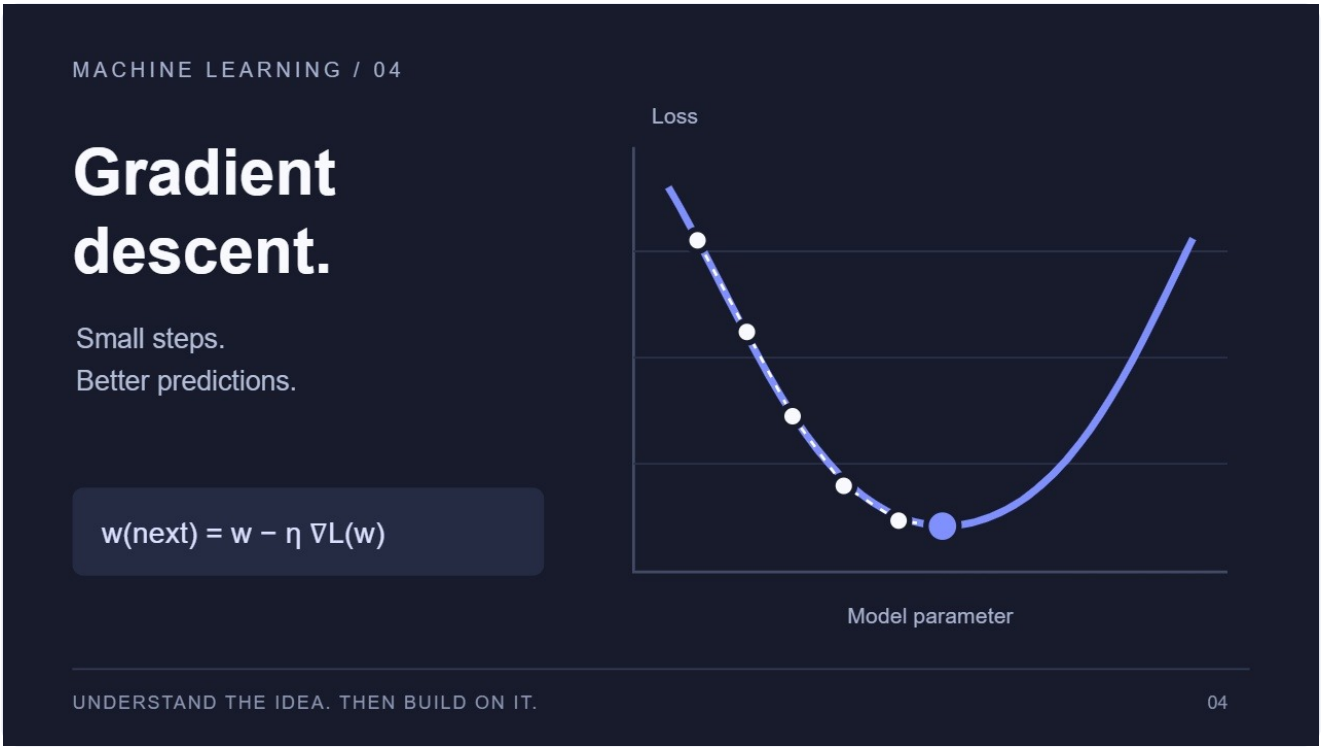
Choose a useful step size

Move against the gradient, then check the resulting loss. Controlled steps can approach a minimum; overly large updates may overshoot and become unstable.

KEY POINTS

- Move opposite the gradient to reduce the loss locally.
- The learning rate determines how far each update moves.
- Use the loss trend to judge whether the updates are improving the model.

NOTES & SCREENSHOTS



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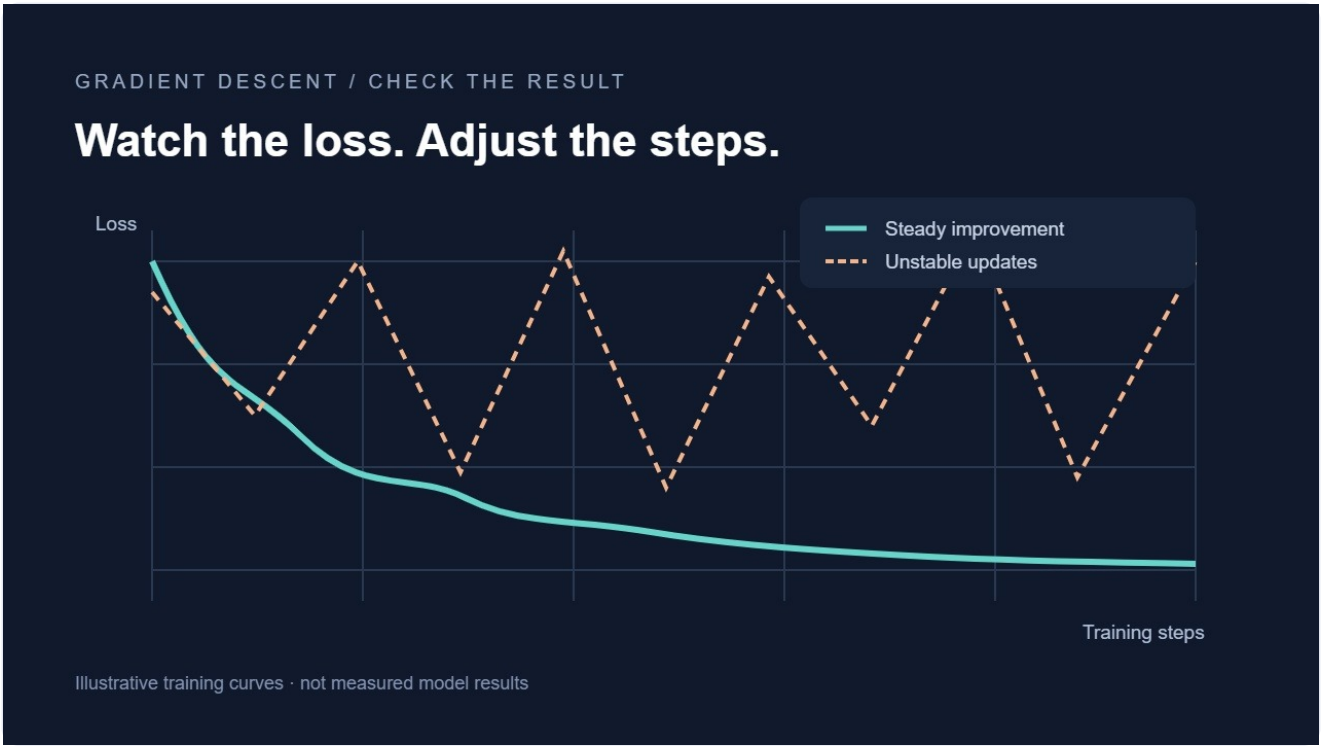
> The gradient gives the direction of steepest increase. To reduce loss, update the parameters in the opposite direction.



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> The learning rate controls the step size. A rate that is too large can overshoot; a very small rate may improve the model slowly.



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> Check the loss across several updates. If it oscillates or grows, try a smaller learning rate and compare the trend.